Interaction in Transformer for Change Detection in VHR Remote Sensing Images

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***Abstract*— With the development of deep learning (DL), very high-resolution remote sensing (RS) image change detection (VHRCD) methods are becoming more popular. However, most existing change detection methods are not good at processing edge details and small target detection. To this end, in this article, InterFormer, a bidirectional interactive framework based on Transformer, is proposed to find small changes and extract more accurate edge information of the change area. First, we designed an asymmetric interaction attention module (IAM) to identify the edge details for the bitemporal image. The IAM fully leverages the benefits of self-attention, performing feature fusion during feature extraction. This approach improves edge feature extraction capability and reduces the number of parameters, compared to other Transformer methods. Second, we designed a global attention-based feature fusion module called global feature fusion module (GFFM) to enhance the detection performance of small targets. The GFFM further improves small target detection ability by augmenting the network's selectivity to spatial information during feature fusion. The method applies to scenarios involving small changes and possesses enhanced edge-detection capabilities. Our method outperforms state-of-the-art counterparts on three public benchmarks and has fewer parameters.**

***Index Terms*— Change detection, global feature fusion module (GFFM), interaction attention module (IAM), InterFormer, Transformer.**

I. Introduction

REMOTE SENSING (RS) change detection, commonly referred to as CD, is a technique used to identify differences in remote sensing images (hereafter referred to as the bitemporal change detection method) captured at different times [1]. It plays a key role in various aspects of human development, such as urban change detection [2], disaster level assessment [3], desertification change detection [4], environmental change detection [5], and land-use rate detection [6]. Recently, numerous researchers have investigated diverse CD methods based on deep learning (DL), which has fostered continuous innovation in applied remote sensing [7].

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In recent years, progress in DL methods has led to the development of various deep neural networks (DNNs), such as deep belief networks (DBN) [8], multilayer perceptrons (MLP) [9], deep convolutional neural networks (CNNs) [10], and transformers [11]. These networks can autonomously learn abstract and multilevel features of complex bitemporal images and exhibit strong robustness against noise. They are employed for change detection in bitemporal remote sensing images. Although these models provide faster and more accurate results compared to traditional detection methods, they still face challenges such as high computational complexity and complicated network design [12]. Fully convolutional networks (FCNs) [13] have been proposed to address these drawbacks. FCNs eliminate the need for a fully connected layer, which can still result in information loss of bitemporal images due to the direct convolution approach. To tackle this challenge, researchers have employed the dual-branch network, originally introduced by the Siamese network (for instance, deeper semantic information and information between bitemporal images) to enhance detection accuracy significantly. Jaturapitpornchai et al. [14] proposed a network based on U-Net to detect changes in buildings within the same area from two synthetic aperture radar (SAR) images acquired at different times [14]. Hamdi et al. [15] employed an enhanced U-Net network to detect affected areas in the Arc Geographic Information System (ArcGIS) environment, utilizing data for model training derived from forest areas in Bavaria, Germany [15]. Peng et al. [16] proposed an improved network based on UNet++ that enables end-to-end change detection for very high-resolution (VHR) satellite images [16]. The authors incorporated complex hop connection structures at different network layers to extract differential features from images better. Furthermore, they applied a residual block strategy to accelerate the gradient convergence of the network. The spatiotemporal attention network (STANet) [17] leverages channel information, spatial information, and relationships between similar objects at different times and locations within the image to introduce attention to change detection, thus becoming a part of the image feature extractor. Difference enhancement and spatial–spectral nonlocal network (DESSN) [18] introduces a lightweight difference enhancement and nonlocal spatial spectrum information fusion method for VHRCD, which can amplify the differences in bitemporal images while maintaining the edge integrity of objects.

With the development of the Transformer, various researchers have extensively investigated RS image change detection. Chen et al. [23] proposed the BiT model, which

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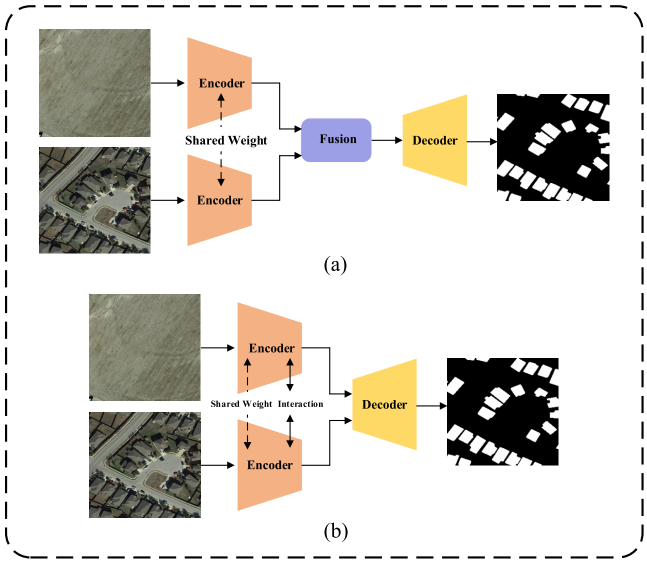


Fig. 1. Image depicts two distinct strategies. (a) Extraction–fusion strategy. (b) Extraction–fusion–extraction strategy.

represents bitemporal images as multiple labels and employs a transformer encoder to model the context within a compact, label-based space–time [19]. Subsequently, the learned context-rich labels are fed back into the pixel space to refine the original features through the transformer decoder. Bandara et al. [20] designed ChangeFormer, which utilizes a hierarchical transformer encoder to extract both coarse and fine features from bitemporal images, followed by four feature difference modules that calculate feature differences at various scales [20]. Ultimately, a lightweight MLP decoder combines multilayer feature differences to predict the mask. Zhang et al. introduced SwinSUNet [21], which is a U-shaped structure that incorporates a Siamese network with a Swin Transformer [22].

In recent years, many change detection methods have adopted the extraction-fusion strategy, as shown in Fig. 1(a). These methods include STANet, BiT, ChangeFormer, and SwinSUNet. However, methods that incorporate additional fusion during the feature extraction process exist to enhance the capture of differential characteristics in bitemporal images, as illustrated in Fig. 1(b), such as DESSN. Despite the numerous Transformer models proposed and utilized in VHRCD, specific issues persist in these Transformer-based networks [23]. First, existing methods primarily employ the Transformer as a feature extractor. Although these methods have achieved satisfactory results, it is still necessary to thoroughly consider the semantic information of closely linked targets and those with subtle changes. This shortcoming hinders the effective establishment of differences between bitemporal images, thus affecting the detection outcomes. Second, the current Transformer networks increase the scale and computational cost of the model.

To address these challenges, this article proposes a bidirectional interactive framework based on a Transformer called InterFormer. This framework divides the bitemporal images into equal-sized vectors for input. The IAM employs an asymmetric feature extraction module (FEM) that concurrently extracts features from the bitemporal images and performs fusion. A pyramid structure is formed by superimposing the IAM, facilitating the extraction of multilayer features. More specifically, larger calculation matrices will be generated when the multitemporal image size is larger. Therefore, we designed a simple preprocessing module to expand and increase the size. In addition, using the same positional encoding for bitemporal images also improves performance. For the multilayer extracted feature maps, a feature fusion combined with global attention is constructed to allow robust fusion of different feature maps and change information at different scales.

In summary, the main contributions of this work are as follows:

1) We designed a Transformer framework called InterFormer, featuring asymmetric attention, which can effectively learn the features of bitemporal images and automatically extract the distinct parts from them.
2) We introduced the IAM, which includes the FEM and the feature interaction module (FIM). FEM employs the concept from the Siamese network for feature extraction in bitemporal images. FIM directly interacts with bitemporal images for feature fusion and improved feature extraction. The next stage of FEM benefits from better feature extraction through the previous stage of FIM. This approach aims to leverage the advantages of self-attention and address the issue of the Siamese network not interacting directly. Additionally, the extraction–fusion–extraction method better targets edge details, resulting in higher precision and reduced network parameters.
3) We designed the global feature fusion module (GFFM), which can directly connect features in different dimensions, effectively correlate them, and focus on the similarity between the two to enhance the detection performance for targets of varying sizes.

II. Related Work

A. RS Image Change Detection Methods

Change detection evaluates the differences by examining bitemporal images generated from the same area at different periods. RS image change detection involves analyzing multisource remote sensing images and related regional spatial data from the Earth's surface at various times [24]. By utilizing the characteristics of associated features and remote sensing imaging mechanisms, photography, graphic information processing technology, and mathematical simulation techniques are employed to investigate and study the geographical changes in the region [25]. These changes may include the number of features, regional shifts, and feature characteristics and status alterations. The primary objective of this research is to identify the most sensitive change signals and eliminate disparate signals as influencing factors.

In the past decade, the rapid development of big data analysis and artificial intelligence has brought the field of change detection to new heights [25]. Due to advancements in remote sensing big data analysis and artificial intelligence, deep-learning methods have quickly been incorporated into the application field of remote sensing image change detection.

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As an effective technique for mining information features from high-resolution remote sensing data, deep-learning methods offer a new path for feature analysis and change detection in high-resolution image data [26]. Deep-learning methods can automatically and comprehensively obtain many abstract features of complex ground objects, proving an efficient feature-learning method [6]. Abstract features obtained by deep networks exhibit excellent robustness to noise systems and can also handle many homologous and heterologous, as well as multitemporal remote sensing image data [6]. Jat proposed a network based on U-Net [14] to detect changes in buildings in the same area from two SAR images taken in different periods. Hamdi used an improved U-Net system to detect affected areas under ArcGIS conditions, with the model training data primarily sourced from the forest areas of Bavaria, Germany. STANet [17] utilized channel information, spatial information, and the relationship between similar objects at different times and locations of the image to introduce attention to change detection, becoming a part of the image feature extractor. This approach designs two distinct attention mechanisms to enhance detection accuracy and reduce false-positive occurrences. However, its efficacy declines when dealing with areas of minimal color variations. Tao proposed a remote sensing image differential enhancement and spatial spectrum nonlocal network, improving U-Net by enhancing the difference of bitemporal images [18]. This method enhances the difference between bitemporal images and minimizes the influence of irrelevant changes, yet it still has room for improvement in the detection of small targets. Due to their excellent speed and accuracy, the U-Net series has become the mainstream RS image change detection algorithm. With the development of the Transformer, related methodologies have been applied to change detection. Gedara's ChangeFormer [20], which combines a hierarchical Transformer encoder with four differential feature modules, is used to extract fine and coarse features from bitemporal images at various scales, demonstrating exceptional performance on multiple datasets. However, this method involves numerous parameters and lacks focus on edge detail information.

Zhang et al. designed SwinSUNet [21], a pure Transformer change detection network that leverages the Swin Transformer's efficient feature extraction capability. Although it augments edge detail extraction ability, the network is marked by several limitations such as high computational burden and lack of support for small targets. Most deep-learning-based methods employ Siamese network algorithms. However, these models have only been used to detect simple land cover changes, and these algorithms need to be more robust for complex edge information and small targets [27].

B. Transformer

The Transformer is a deep neural network employing self-attention [28]. Initially developed for natural language-processing tasks, it typically comprises an encoder module and a decoder module [29]. The model is constructed using a series of encoders and decoders with similar architecture. Each encoder consists of a self-attention layer and a feedforward neural network, while each decoder features a self-attention layer, an encoder–decoder, and a feedforward neural network. Before using the Transformer for language translation, the vocabulary of all languages is transformed into vector embeddings. Owing to the Transformer's impressive representational capabilities, researchers have proposed extending its application to machine vision tasks. Similar to other network models, such as convolutional and recurrent networks, the Transformer demonstrates competitive or superior performance across various visual benchmarks [30].

Generative image pretraining methods have significantly evolved since their inception. Chen et al. [23] revisited this approach, integrating recent advancements in self-supervised learning techniques. The training process comprises two phases: pretraining and fine-tuning. During the pretraining phase, the autoregressive bidirectional encoder representation from the transformers (BERT) method is predominantly employed. Dosovitskiy et al. introduced a streamlined transformer, also known as the vision transformer (ViT) [31], which exhibits improved performance in visual recognition tasks when directly applied to image block sequences. The model design closely follows the original transformer. ViT segments the input image into equally sized patches, transforming each patch into a fixed-length token and subsequently inputting it into the multihead attention. The multihead attention calculates the attention weight of a Query token which could be the prompt of an image. Both the key (K) token and the value (V) associated with each K are multiplied together. We can also say it calculates the relationship or attention weight between the query (Q) and the K and then multiplies the V associated with each K. The principles governing the subsequent encoders align with those of the original transformer.

However, due to the classification of images, a unique token is added to the sequence, and the output corresponding to this token serves as the category prediction. Instead of merely incorporating the Transformer and adding functionalities to existing architectures, Carion further designed a dedicated object detection model, providing a detection transformer (DETR) [32]—a simple and comprehensive end-to-end object detector. By appending a set of mask heads to the decoder, DETR can be naturally extended to panoptic segmentation tasks, yielding more valuable data. Wang et al. introduced max-deeplab technology for prediction, leveraging the remarkable segmentation results of the mask transformer without replacing subtasks such as bounding box detection. Wang et al. also proposed a Transformer-based video instance segmentation method called VisTR [33], which takes an image sequence as input and generates relevant pose estimation results. An effective matching method is provided, establishing ground truth for the prediction outcomes.

C. Attention Mechanisms in Computer Vision

The attention mechanism, which enables the network to focus on the most critical areas of an image while disregarding unrelated information, was inspired by humans' ability to understand and analyze complex scenes through their eyes and brains [34]. The human brain can concentrate on areas of interest, and researchers have designed attention to emulate

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this ability. In the visual system, the attention adaptively weights the network based on the input region's importance, performing differential attention [35].

In computer vision, reducing image dimensions and increasing the number of image channels are commonly employed techniques for effective feature extraction. Different channels correspond to distinct characteristic targets. Channel information can be viewed as selecting essential targets, enabling the network to choose crucial semantic information adaptively. In this context, Hu et al. [36] proposed the channel attention, SENet, which aims to gather information between various channels and establish relationships through a squeeze-and-excitation (SE) block. However, SENet solely relies on the global average pooling to capture global information, which constrains its ability to model target features. To address this issue, Gao et al. [37] introduced an enhancement to the SE module that employs the global second-order pooling (GSoP) block to establish depth feature information while collecting global data. Considering the computational cost and parameter count of fully connected SE modules, incorporating one SE block after each convolution layer results in significant time and space overhead. Moreover, using the fully connected layer to establish connections between channels is a complex process. To address the issues, Yang et al. [38] introduced gated channel transformation (GCT) to collect valuable high-dimensional information and explicitly model the relationships between channels. Spatial attention can be regarded as a selection mechanism among different features within a single adaptive channel. For large-sized images, dimensions are typically reduced through convolution layers, which facilitates the fusion of in-depth information pertaining to the same object. However, this approach does not establish relationships between objects. To concentrate limited computing resources on essential targets, Mnih et al. [39] proposed a recurrent attention model (RAM), which employs RNNs to enable the network to learn where to focus. SKNet [40] integrates multiple receptive field feature maps to apply attention to receptive fields. By training the network weights, targets of varying sizes will extract target features through a single receptive field and acquire these features by superimposing the adaptive weights of different receptive fields. Channel and spatial attention integrate the benefits of both channel and spatial attention. Woo et al. introduced two hybrid attention mechanisms. BAM, which focuses on the influence of attention on general DNNs, proposed a simple, yet effective attention model that can be incorporated into any feedforward convolutional neural network [41]. This model derives an attention map through two separate paths: channel and spatial. The hybrid attention mechanism, convolutional block attention module (CBAM) [42], combines channel attention with spatial attention. This approach decouples the channel attention map and the spatial attention map, enhancing computational efficiency.

III. Methods

A. Overview

In this section, we propose an end-to-end network InterFormer based on the Transformer to enhance the ability to extract edge details and small targets of CD methods. The proposed method framework is shown in Fig. 2, which consists of a dual-branch interactive encoder and a feature fusion module. The framework includes two main modules: IAM and GFFM. First, the bitemporal image is input into a multilayer network stacked by IAM for feature extraction and feature fusion. Since the input image is 256×256, the direct input of the IAM module will generate a complex calculation matrix, so we designed a preprocessing head to achieve the scaling and dimension increase. The preprocessing module is inspired by the fact that the deformable convolution is better than the convolution module for irregular size changes [43]. Therefore, a simple self-attention can be implemented in parallel. Then, the feature maps of different sizes after fusion are generated by three-layer IAM. GFFM is a feature fusion module based on global attention, which is used to enhance the receptive field of small targets and fuse feature maps of different sizes. Finally, the image size is restored to 256×256, and the result is output after normalization. According to the threshold setting of the loss function, the value of each pixel can only be represented by 1 and 0. The binarization process is defined as follows:

$$Y_{i,j} = \begin{cases} 0, & 0 \le P_{i,j} \le T \\ 1, & T \le P_{i,j} \le 1 \end{cases} \tag{1}$$

where $Y_{i,j}$ and $P_{i,j}$ stand for the value of the ith row and the jth column before and after binarization, respectively. T represents the threshold we set, which is usually set to 0.5.

B. IAM

Generally, CD methods based on the Transformer directly employ the Transformer for feature extraction, where Q, K, and V act on an image. This feature extraction method solely uses the Siamese network's shared weight to extract the same or different features. This approach, however, needs to fully leverage the interactive capability of self-attention. Consequently, our IAM utilizes positional encoding and concatenation to input the dual-time images into FEM for feature extraction and employs FIM for feature interaction. The advantage of this technique is that the network can simultaneously extract and fuse features. Additionally, to prevent the bitemporal image background from influencing the foreground, we only perform interactions between the k vector and the v vector to enhance fusion efficiency and separate the background in both instances, as shown in Fig. 3.

Input_A and Input_B through the preprocessing head are input into FEM separately, and the dimensions are scaled through a convolution and layer normalization and then input into self-attention. Using the idea of a Siamese network, the bitemporal images share the weight of the module for the first feature extraction. Specifically, the input of the multihead attention is divided into three branches, namely Q, K, and V. The input bitemporal image feature sequence can be independently learned to obtain h groups of different linear projections to transform Q, K, and V. Then, this set of transformed queries, keys, and values will be sent to attention aggregation in parallel.

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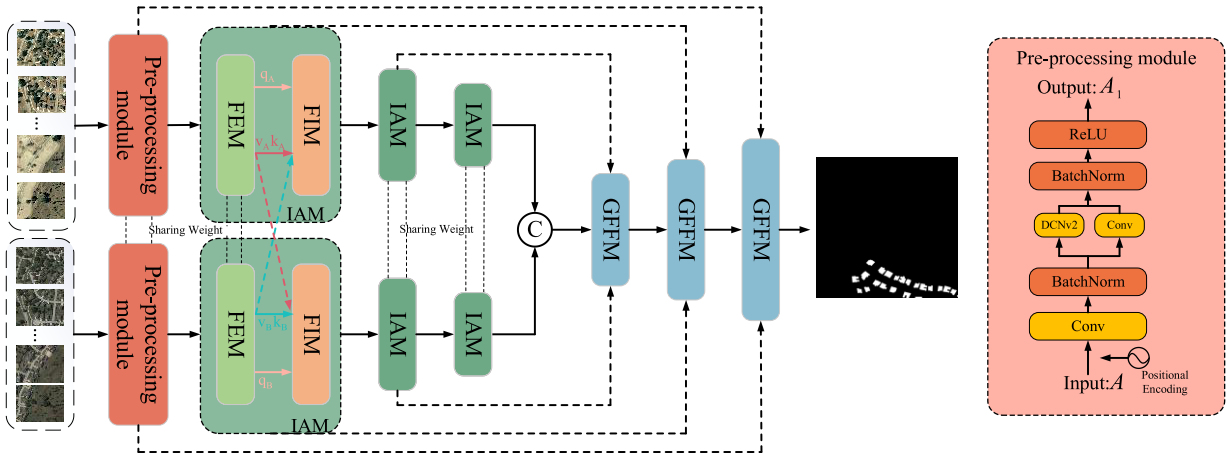


Fig. 2. InterFormer is an end-to-end model. The network structure has only two modules. One is a backbone network based on the IAM proposed in this article to extract and fuse features and then connect GFFMs for fusion. The network framework adopts U-Net. As the network deepens, the resolution gradually becomes smaller, and the number of channels gradually increases.

Attention is the basic element of the design feature fusion network. Given the Q, K, and V, the attention function is the scale dot product attention, defined in the following equation:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \tag{2}$$

where d_k is the scaling factor.

Single attention can only focus on the distribution is not comprehensive, so it is necessary to make multiple overlays to adapt to different change areas so that the model focuses on different information.

$$\text{MultiHead}(Q, K, V) = \text{Concat}(H_1, \ldots, H_h)W^O \tag{3}$$

$$H_i = \text{Attention}\left(QW_i^Q, KW_i^k, VW_i^V\right) \tag{4}$$

where $W_i^Q \in R^{d_m \times d_k}$, $W_i^K \in R^{d_m \times d_k}$, $W_i^V \in R^{d_m \times d_k}$, and $W_i^O \in R^{n_h d_v \times d_m}$ are parameter matrices. h is the number of attention heads.

$$Q_A = K_A = V_A = \sigma(\text{LayerNorm}(\text{Conv}(\text{Input}_A))) \tag{5}$$

$$Q_B = K_B = V_B = \sigma(\text{LayerNorm}(\text{Conv}(\text{Input}_B))) \tag{6}$$

where $\sigma(\cdot)$ denotes the multihead attention.

To effectively fuse the features of the bitemporal images, we employ an asymmetric hybrid attention strategy. This is because utilizing cross-attention increases the weight of negative samples, which is not favorable for extracting differential features. To reduce the computational complexity of the network, we only use K and V for cross-attention fusion to enhance the understanding of differential features. The specific formula is

$$\text{Attention}_A\left(Q_A, K_{A,B}, V_{A,B}\right) = \text{softmax}\left(\frac{Q_A K_{A,B}^T}{\sqrt{d_k}}\right)V_{A,B} \tag{7}$$

$$\text{Attention}_B\left(Q_B, K_{A,B}, V_{A,B}\right) = \text{softmax}\left(\frac{Q_B K_{A,B}^T}{\sqrt{d_k}}\right)V_{A,B} \tag{8}$$

where d represents the dimension of the key, while Attention_A and Attention_B establish feature maps of both before and after the change. They incorporate both self-attention and cross-attention for feature extraction and feature fusion separately.

Finally, we use a simple MLP module to transform the semantic space. The specific formula is as follows:

$$\text{output} = \text{MLP}(\text{LayerNorm}(\text{CrossAttention}(x))). \tag{9}$$

Utilizing the IAM module offers several advantages over traditional transformer networks. First, conventional methods obtain different information through feature extraction, which is less efficient and needs more direct interaction between bitemporal images. By employing the IAM module, the interaction between bitemporal images is realized by establishing connections between channels, directly enhancing the network's recall rate. Second, the network does not automatically focus on differences when using a Siamese network as the backbone. The attention mechanism can automatically select where the network is interested (areas with differences), which is a significant advantage of using attention. Third, as the network depth increases, the Transformer may lose some semantic information related to small targets. The IAM module assists the network in extracting small target information through an iterative method of feature extraction, fusion, and extraction, ensuring the network's precision.

C. GFFM

Most U-Net-based feature fusion layers are implemented using max-pooling and convolution modules. This approach allows the deep network to obtain deep semantic information about the target while neglecting the fine-grained features of small targets. To address this issue, the FPN is designed. However, due to the substantial depth of the U-Net, this additional upsampling layer may overwhelm the characteristics of small targets, leading to suboptimal detection performance. Consequently, we propose a simple and efficient fusion strategy to overcome this limitation, as shown in Fig. 4.

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First, deconvolution is employed to increase the dimensionality of depth features, followed by the concatenation of the preceding layer's features. The spatial attention is then utilized to transform the fused features semantically, combined using a residual connection similar to ResNet [44]. This approach is beneficial for representing shallow feature information and the fusion of in-depth information. Finally, channel attention is applied to extract the fused information once more, and a convolutional layer is concatenated to enable the conversion of channel numbers. The process can be expressed using the following formula:

$$\text{OutPut} = dp(\text{Global}(\text{TransConv}(x_i), x_{i-1})), dp(x_{i-1})) \quad (10)$$

where dp means dropout.

IV. Experiments

A. Datasets

1) LEVIR-CD Dataset: The LEVIR Building Change Detection dataset is a new large-scale remote sensing building change detection dataset provided by Beijing Airlines [45]. This dataset comprises 31 333 examples for training and detection, consisting of 637 ultrahigh-resolution image blocks sourced from Google Earth, each with dimensions of 1024 × 1024. The spatial resolution of the images is 50 cm/px. LEVIR-CD covers various building types, including building areas, large warehouses, residential buildings, and small garages. The dataset primarily focuses on changes related to facilities, encompassing building growth (from soil, grass, hardened ground, or buildings under construction to new building areas) and building decay.

2) CDD Dataset: The dataset was released in 2019 and comprises 35 types of remote sensing features and 224 bands [46]. It includes static composite images of the target, composite images of the target with minimal changes, and actual remote sensing images of seasonal changes obtained from Google Maps. The seasonal remote sensing images have a size of 256 × 256 and encompass 16 000 image sets, including 10 000 training sets and 3000 test and validation sets. The spatial resolution ranges from 3 to 100 cm/px.

3) DSIFN Dataset: The dataset was manually collected from Google Earth [47]. It has an image size of 512 × 512, and after data enhancement, a total of 3940 bitemporal image pairs were obtained by dividing five image pairs into 394 subimage pairs. Xi's image pairs were further divided into 48 image pairs for model testing. The dataset contains 3600 image pairs for training, 340 pairs for validation, and 48 pairs for testing. It includes images from six cities in China: Beijing, Chengdu, Shenzhen, Chongqing, Wuhan, and Xi'an.

B. Experimental Setup

1) Datasets: This article takes public datasets for experiments, namely the LEVIR-CD dataset, the CDD dataset, and the DSIFN dataset.

2) Evaluation Indicators: We select the three most used evaluation indicators for RS image change detection to comprehensively evaluate the experimental results, which are precision (Pre), recall (Rec), F1-score (F1), and IOU.

a) Precision: Precision, or precision, reflects the ability of a classifier or model to predict the accuracy of a positive sample correctly, that is, how many of the predicted positive samples are real positive samples. The formula is

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \quad (11)$$

b) Recall: Recall, also known as the recall rate, is the number of all positive class pairs, reflecting the ability of the classifier or model to correctly predict the fullness of positive samples, increasing the proportion of positive samples predicted as positive samples, that is, positive samples are predicted as positive samples in the total positive samples. The formula is

$$\text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (12)$$

c) IOU: The meaning of the intersection and union ratio is the ratio of the intersection and union of the model's prediction result and the real value of a certain category. For change detection, the intersection over union between the predicted mask and the real mask is calculated.

$$\text{IOU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \quad (13)$$

d) Truth positive (TP): Values that are actually positive and predicted positive; Truth negative (TN): Values that are actually negative and predicted to be negative; False positive (FP): Values that are actually negative but predicted to be positive; False negative (FN): Values that are actually positive but predicted to negative.

e) F1-Score: The precision rate and recall rate are interconnected, and ideally, both should be high. However, they tend to be mutually restrictive, as pursuing high precision often leads to a low recall rate, and aiming for a high recall rate can adversely affect accuracy. It is essential to consider precision and recall rates holistically since they can sometimes be contradictory. The F-score, a commonly used method, provides a more comprehensive evaluation of a classifier by balancing the effects of accuracy and recall. The F1-score, the harmonic mean of precision and recall, indicates a model's quality, with a larger F1-score signifying a superior model.

$$\text{F1-score} = (1 + \beta^2)\frac{\text{Pre} \cdot \text{Rec}}{\beta^2(\text{Pre} + \text{Rec})}. \quad (14)$$

Here, β is used to balance Precision, and Recall's weight in the F-score calculation. If 1 is taken, Precision is as essential as Recall; if less than 1, Precision is more critical than Recall; if greater than 1, Recall is more important than Precision.

3) Loss Function: RS image change detection primarily involves pixel-level segmentation detection algorithms, which can be categorized into two groups: change pixels (denoted by 1) and unchanged pixels (denoted by 0). In this article, the cross-entropy loss is employed as the loss function, effectively integrating log-softmax and nll loss into a single class.

The definition of log-softmax is as follows:

$$f_i(x) = \log \frac{e^{(x_i)}}{a}, \quad a = \sum e^{(x_i)}. \quad (15)$$

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The definition of nll loss is as follows:

$$\text{loss}(x, \text{class}) = -x[\text{class}]. \tag{16}$$

The target expected by this loss is the index of the category (0 to $N-1$, where N = number of classes).

The definition of cross-entropy loss is as follows:

$$\text{loss}(x, \text{class}) = -\log \frac{\exp(x[\text{class}])}{\exp\left(x\left[j\right]\right)}. \tag{17}$$

4) Implementation Details: We implemented the proposed method with the PyTorch framework and trained it on a GeForce RTX 3090Ti GPU. The number of training cycles is set to 500. The optimizer selects Adam. During training, the learning rate is adjusted according to the set interval. The initial setting is 0.0001, and the learning rate begins to decay after 400 rounds. The batch size is set to 8.

5) Parameter Settings: Our method allows setting parameters in two parts: one influences the network width, while the other affects the network depth. Expressly, it accepts an input that is an integer multiple of scale 16 and can modify the output by altering the number of layers in the IAM. Within the IAM, parameters such as embedding and parallel attention heads can be adjusted. However, it is crucial to ensure that parallel attention heads are divisible by the dim of embedding as an integer.

C. Comparison Experiments

1) To Demonstrate the Effectiveness of InterFormer in RS Image Change Detection, This Article Compares Its Performance With Three Top-Performing Methods on Three Different Datasets as of November 2022: BiT, STANet, ChangeFormer, DESSN, SwinSUNet, and ECF-Net [48]. BiT incorporates the transformer concept into change detection, utilizing a classification token in a classified ViT to learn integrated transformer features and relying on the token for classification prediction. BiT learns to integrate the truly changing parts of the two images by employing a semantic token. STANet designs a self-attention based on spatiotemporal dependence to model spatiotemporal relationships and adds a CD self-attention module during feature extraction to generate more distinct features. ChangeFormer combines a hierarchical transformer encoder with a multilayer perceptron (MLP) decoder in a Siamese network structure, effectively capturing the multiscale remote details necessary for accurate change detection. DESSN introduces a lightweight difference enhancement and nonlocal spatial–spectral information fusion method for ultrahigh resolution remote sensing image change detection, addressing the issue of low tightness within changing targets, refining edge information, and significantly reducing the computational complexity of the proposed model. SwinSUNet identifies the limitations of CNN-based methods in extracting long-range global features due to the inherent locality of convolution operations, exploring the potential of pure Transformer networks in change detection tasks. ECF-Net enhances the utilization of fine-grained information in multiscale feature maps for result prediction by constraining the number of feature channels in the fusion process through a feature comparison module. For

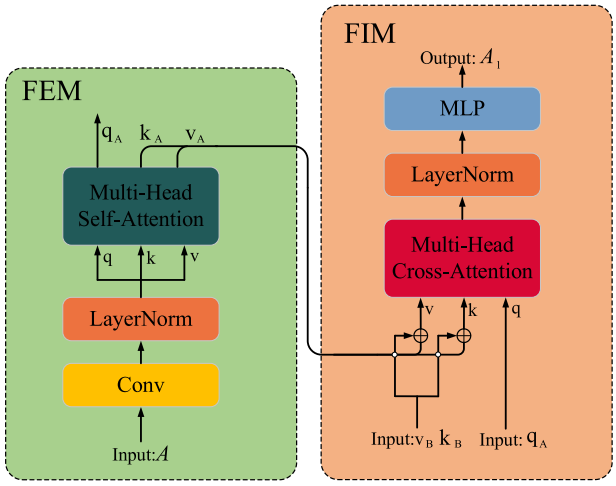


Fig. 3. IAM comprises FEM and FIM. The bitemporal images share the network weight of IAM. FEM employs Conv for scaling, followed by self-attention for feature extraction. FIM stitches the K and V dimensions of the bitemporal images, utilizes Q to extract the distinct features of the bitemporal images, and ultimately outputs through a shared MLP.

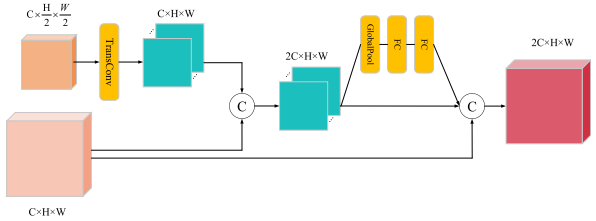


Fig. 4. GFFM module. The multiscale output of the IAM pyramid structure is taken as the input. It is used as feature fusion to connect layer x_i and layer x_{i-1} every time.

TABLE I
QUANTITATIVE ANALYSIS OF THE LEVIR-CD DATASET CD RESULTS. THE BEST VALUES ARE IN BOLD

Methods	Pre(%)	Rec(%)	F1(%)	IoU(%)
W-Net [49]	81.01	87.17	83.91	
STANet [17]	83.81	91.00	87.26	77.40
DSIFN [47]	86.00	89.73	87.83	
BIT [19]	89.24	89.37	89.31	80.68
ChFormer [20]	92.05	88.80	90.40	82.48
DESSN [18]	90.99	91.73	91.36	83.89
ours	93.22	**91.76**	**92.69**	**86.22**

Figs. 5–7: for the convenience of viewing, we use different colors to express different meanings. White means true positive, black means true negative, red means false positive, and blue means false negative.

2) Experiment on the LEVIR-CD Dataset: We compared our model with eight popular models, and the experimental results are presented in Table I, with the best values highlighted in bold. We selected the top three performing models for visualization, as depicted in Fig. 5. We examined three types of targets: large targets, dense small targets, and targets with minor differences. The results indicate that both (d) and (e) failed to detect large targets, with the edge in the first row

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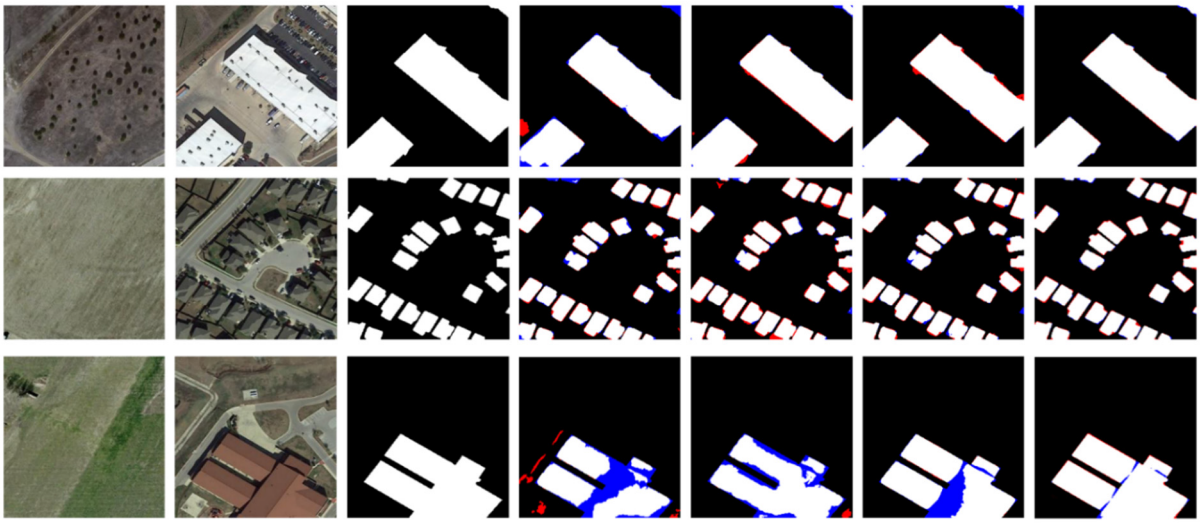


Fig. 5. Visual comparison of results of three different situations. Large target (row 1), small target (row 2), and dense scene (row 3). (a) Pretemporal image. (b) Posttemporal image. (c) Label. (d) BiT. (e) ChangeFormer. (f) DESSN. (g) Ours.

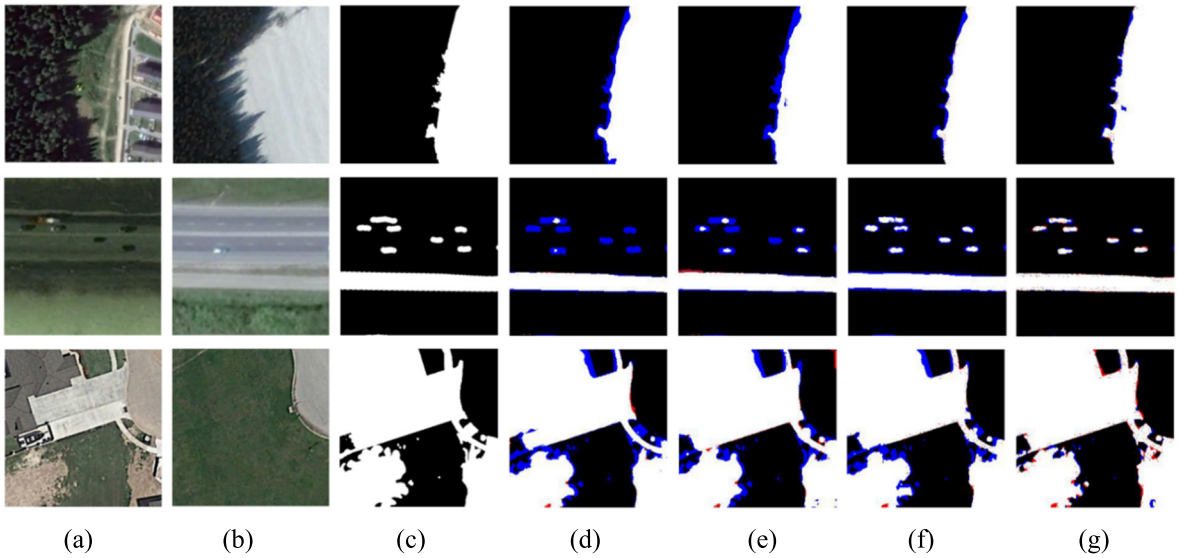


Fig. 6. Visual comparison of results of three different situations. Large target (row 1), small target (row 2), and dense scene (row 3). (a) Pretemporal image. (b) Posttemporal image. (c) Label. (d) DESSN. (e) STANet. (f) SwinSUNet. (g) Ours.

not displayed. Although (f) and our model revealed the edges, our model demonstrated more terrific refinement. For dense small targets, (d) exhibited incomplete detection and adhesion, while our model outperformed (e) and (f) in terms of accuracy for small target detection. In the case of minor differences, as illustrated in the third row, light changes and occlusion led to false detections in (d)–(f). However, our model showcased the most sophisticated detection, emphasizing its effectiveness.

3) Experiment on the CDD Dataset: Our algorithm was compared with six popular algorithms, and the experimental results are presented in Table II, with the best values highlighted in bold. Our InterFormer demonstrated strong performance across various metrics, with its F1 score being 3% higher than the current leading SwinSUNet algorithm. We selected the three best-performing models for visual comparison with our model, as depicted in Fig. 6. We examined three types of targets: large targets, dense small targets, and targets with minor differences. Upon comparing the first row of images, we observed that (d) exhibited inaccuracies in extracting small target information, while (f) lost some small target details. Analyzing the second row of images revealed that both (e) and (f) had adhesions and could not fully extract edge information. In the third row of images, (d) displayed some false detections, and (e) extracted information with limited accuracy. By contrast, our model demonstrated superior detection capabilities and increased robustness across multiple datasets.

4) Experiment on the DSIFN Dataset: Our algorithm was compared with six popular algorithms, and the experimental results are presented in Table III, with the best values highlighted in bold. It was observed that our InterFormer outperformed the others across all metrics. We selected the

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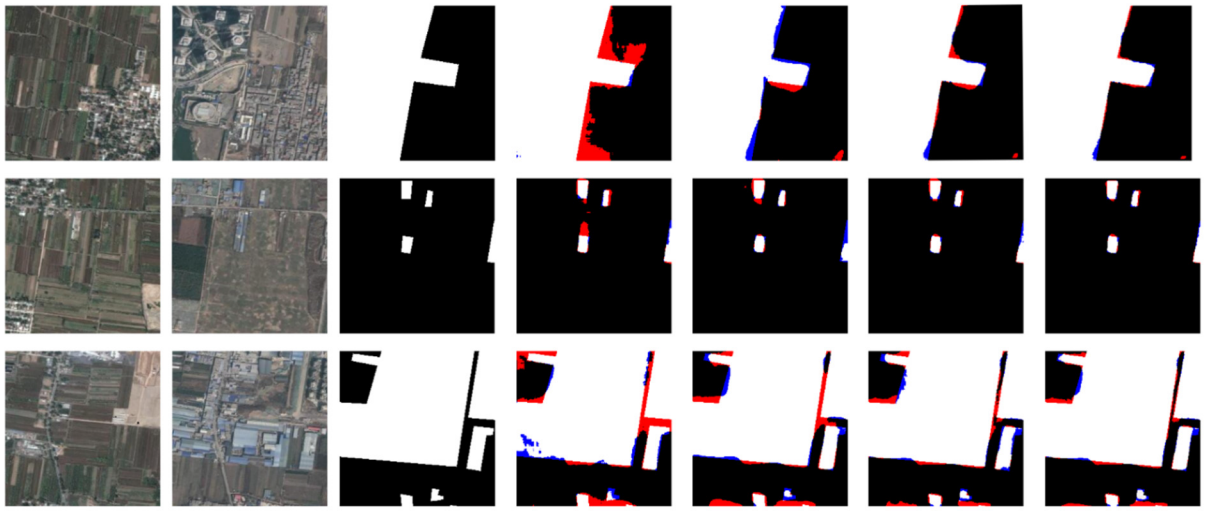


Fig. 7. Visual comparison of results of three different situations. Large target (row 1), small target (row 2), and dense scene (row 3). (a) Pretemporal image. (b) Posttemporal image. (c) Label. (d) ChangeFormer. (e) DESSN. (f) ECF-Net. (g) Ours.

TABLE II
QUANTITATIVE ANALYSIS OF THE CDD DATASET CD RESULTS. THE BEST VALUES ARE IN BOLD

Methods	Pre(%)	Rec(%)	F1(%)	IoU(%)
Siam-UNet [16]	88.31	85.30	86.78	-
DSIFN [47]	90.72	86.50	88.56	-
DESSN [18]	95.04	88.77	91.80	85.19
STANet [17]	93.10	93.90	93.51	88.94
SwinSUNet [21]	95.70	92.30	94.00	91.27
ours	**97.08**	**97.01**	**97.05**	**93.97**

TABLE III
QUANTITATIVE ANALYSIS OF THE DSIFN DATASET CD RESULTS. THE BEST VALUES ARE IN BOLD

Methods	Pre(%)	Rec(%)	F1(%)	IoU(%)
STANet [17]	67.71	61.68	64.56	47.66
BIT [19]	68.36	70.18	69.26	52.97
ChFormer [20]	88.48	84.94	86.67	76.48
DESSN [18]	94.11	94.03	94.07	88.79
ECF-Net [48]	94.34	94.25	94.29	89.20
ours	**95.18**	**95.56**	**95.36**	**91.87**

three best-performing models for visual comparison with our model, as depicted in Fig. 7. We examined three types of targets: large targets, dense small targets, and targets with minor differences. When comparing the first row of images, we found that (d)–(f) exhibited false detections during the extraction of edge information for large targets, as well as false detections of small targets in the upper right corner. Upon analyzing the second row of images, we observed that both (d) and (e) had adhesions on adjacent targets. In the third row of images, our model demonstrated superior edge detection performance and effectively handled false detections caused by illumination.

TABLE IV
COMPARISON OF THE EFFICIENCY OF DIFFERENT NETWORKS ON THE LEVIR-CD DATASET. THE BEST VALUES ARE IN BOLD

Methods	FLOPs(GB)	Params(MB)	F1(%)
SiamUNet [16]	80.65	39.70	87.11
W-Net [49]	95.80	31.57	83.91
DSIFN [47]	112.15	43.50	87.83
DESSN [18]	36.75	19.35	91.36
ours	8.38	15.10	**92.69**

TABLE V
QUANTITATIVE ANALYSIS OF ABLATION EXPERIMENTS ON THE LEVIR-CD DATASET. THE BEST VALUES ARE IN BOLD

Backbone	FPN	Pre (%)	Rec(%)	F1(%)
Resnet34	U-Net	84.75	89.60	87.11
Resnet34	GFFM	86.75	90.79	88.74
IAM	U-Net	91.87	90.07	90.96
IAM	GFFM	93.65	91.76	92.69

D. Ablation Study

To verify the usefulness of different modules in our InterFormer, we conducted ablation experiments on the LEVIR-CD dataset, including the use of different module combinations. The experimental results are shown in Table IV. As demonstrated in Table V, we replaced the IAM module with ResNet34 as the backbone for our experiments. The results indicated that using the IAM module for feature extraction outperformed ResNet34 in every aspect, with a 3.85% increase in F1 score and more than a 50% reduction in the number of parameters. This is because the IAM module directly extracts

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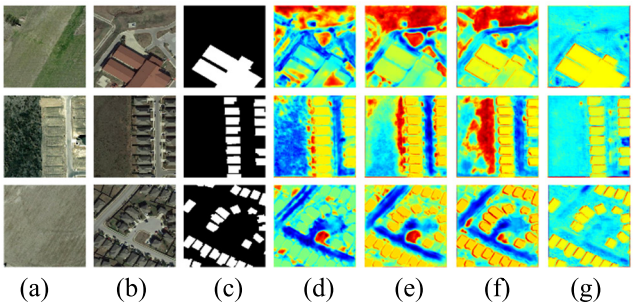


Fig. 8. Visual heat map of output layer CAM. (a) Prechange area. (b) Change map area. (c) Labels. (d) ResNet + U-Net. (e) ResNet + GFFM. (f) IAM + U-Net. (g) IAM + GFFM.

depth features from bitemporal images and establishes their different relationships. In the feature fusion section, the GFFM module strengthens the relationships between different feature maps and enhances their internal cohesion. The GFFM-based feature fusion method surpasses the Siamese-UNet approach, resulting in a 1.63% increase in the F1 score. The combination of IAM and GFFM modules is 5.58% more effective than the native Siamese-UNet, proving the efficacy of our network model.

Fig. 8 presents the visuals we use to examine different modules. The red region represents the network model's area of interest. We display large targets, small targets, and dense targets. Fig. 8(a), (b), and (c) depicts the areas before, during, and after the change, respectively. By comparing (d) and (f) with (e) and (g), we find that using the IAM module enhances the discrimination of small targets while contributing to fine-grained target detection. As shown in Fig. 8(f) and (g), the IAM module provides a more comprehensive detection than Resnet, which can result in large-scale false detections and inaccurate image focus areas. When comparing (d) and (e) with (f) and (g), it becomes evident that employing the GFFM module enables finer edge refinement, ensuring no adhesion occurs when detecting dense targets.

E. Discussion

1) Discussion on Optimal Selection of IAM Module Depth: Since the multihead attention of the IAM module is utilized to extract bitemporal image features, its feature extraction capability directly impacts the model's quality. The experimental results are shown in Table V. We experimented with different depth combinations for feature extraction. The results showed that when the number of multihead attention layers in the IAM module is less than 2, the model loses its feature extraction ability, resulting in a Recall equal to 0. When the number of multihead attention layer overlaps exceeds 8, the model's accuracy does not improve. When the IAM module is iterated four times, we added a superficial downsampling layer at the beginning to maintain the size of the input and output feature maps. We discovered that the model's accuracy and computational complexity are relatively optimal when the combination is [1, 2, 2, 2]. In the case of five iterations of the IAM module, the model's accuracy and computational requirements are relatively optimal when the combination is [1, 2, 2, 2]. When the combination is [1, 2, 4, 8], the model achieves optimal accuracy, but it increases the computational workload, which we believe is not cost-effective.

TABLE VI
COMPARISON OF THE EFFICIENCY OF DIFFERENT MODULE DEPTHS ON THE LEVIR-CD DATASET

Iterations Head	Number Head	F1(%)
4	[1, 1, 1, 1]	null
4	[1, 2, 2, 2]	92.69
4	[1, 2, 4, 8]	93.14
5	[1, 2, 2, 2]	92.76

TABLE VII
COMPARISON OF THE EFFICIENCY OF DATASET SIZE ON THE CDD DATASET

Dataset Size	F1(%)
10 000	88.7
5000	82.3
500	58.9

2) Discussion on Iterative Number of IAM Layers: The number of layers in the IAM module determines the number of subsequent feature fusions. Deeper layers result in a larger receptive field and more extracted depth information. However, as the layers become deeper, the semantic information of small targets may be lost. The number of layers for RS image change detection is maintained at five layers, with feature sizes of (256, 128, 64, 32, 16, 8), where 256 represents the original image size. To explore the impact of different layer depths on the model's performance, we designed experiments with various layers. The experimental results demonstrate that the optimal performance is attained by iteratively applying the IAM for four or five iterations. In this configuration, the $F1$ score reaches an impressive accuracy of 91.78%. When iterating five times, the F1 score increases to 93.15%.

3) Discussion on the Impact of IAM: Our method is inspired by the core idea of ViT, which utilizes the attention mechanism. However, we did not directly adopt the self-attention structure. Instead, we designed a new structure to make it more suitable for the CD approach. The interactive attention operation in our method is more important than directly using self-attention because the CD task's focus is to emphasize the different characteristics in the change area. We conducted a comparative test on the CDD dataset using a noninteractive method and found that the F1 score decreased by 9.8%.

4) Discussion on the Impact of Dataset Size on the Model: Using the CDD dataset as an example, which contains 10 000 bitemporal images for the training set, we train the model without data augmentation and divide it into training sets before testing it on the test set. The experimental results are as Tables VI and VII. This indicates that our model requires more data for training, which is one of the inherent reasons for using the Transformer itself.

5) Discussion on the Model Limitations and Future: Our InterFormer is based on the Transformer model. We have made

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improvements within the Transformer to ensure the overall simplicity and ease of use of the model. This interactive methodology is universally applicable to a wide range of Transformer-based models. Future refinements may prioritize enhancing its computational efficiency while reducing the model size. However, there are some limitations to our method. We did not use a separate module for bitemporal image interaction to maintain a low parameter count. It is mainly manifested in the sensitivity to the detection of shadow changes in buildings. As a result, for more complex detection problems, this method may have limited feature extraction capabilities. To overcome this limitation, the method can be enhanced by adding a separate interaction module for fusion, such as the BiT. By addressing these limitations and continuing to refine the method, our InterFormer model could become even more effective in handling RS image change detection tasks and other complex problems in the future.

V. Conclusion

This article introduces an InterFormer network for VHRCD. The proposed InterFormer network aims to fully leverage self-attention, making detection more concise and efficient. Compared with previous detection methods, our method adopts a direct interaction method in the feature extraction of bitemporal images, which significantly improves the generalization ability of the model for edge and small target detection. By incorporating two modules, the IAM and the GFFM, the primary issues of the prevalent Transformer for remote sensing image change detection are addressed. Specifically, the IAM conducts efficient feature extraction on bitemporal images to better target distinct parts of images. The GFFM facilitates the fusion of IAM features while significantly reducing the number of parameters. Experiments conducted on three popular VHRCD datasets demonstrate the effectiveness of the proposed modules. The results indicate that the InterFormer network outperforms existing networks regarding detection accuracy and efficiency.

The IAM, presented in this article, is based on ViT. However, with the growing popularity of Transformers, the module's design concept can be applied to any architecture incorporating self-attention. Furthermore, experiments were conducted in hyperspectral change detection, revealing that this module can also be applied to feature extraction and fusion of hyperspectral images.

References

[1] C. Huo, K. Chen, K. Ding, Z. Zhou, and C. Pan, "Learning relationship for very high resolution image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 8, pp. 3384–3394, Aug. 2016.

[2] J. Xiao et al., "Evaluating urban expansion and land use change in Shijiazhuang, China, by using GIS and remote sensing," *Landscape Urban Planning*, vol. 75, nos. 1–2, pp. 69–80, Feb. 2006.

[3] Y. Fan, Q. Wen, W. Wang, P. Wang, L. Li, and P. Zhang, "Quantifying disaster physical damage using remote sensing data—A technical work flow and case study of the 2014 Ludian Earthquake in China," *Int. J. Disaster Risk Sci.*, vol. 8, no. 4, pp. 471–488, Sep. 2017.

[4] S. A. Azzouzi, A. Vidal-Pantaleoni, and H. A. Bentounes, "Desertification monitoring in Biskra, Algeria, with Landsat imagery by means of supervised classification and change detection methods," *IEEE Access*, vol. 5, pp. 9065–9072, 2017.

[5] C. A. Mucher, K. T. Steinnocher, F. P. Kressler, and C. Heunks, "Land cover characterization and change detection for environmental monitoring of pan-Europe," *Int. J. Remote Sens.*, vol. 21, nos. 6–7, pp. 1159–1181, Jan. 2000.

[6] L. Khelifi and M. Mignotte, "Deep learning for change detection in remote sensing images: Comprehensive review and meta-analysis," *IEEE Access*, vol. 8, pp. 126385–126400, 2020.

[7] A. A. Aleissaee et al., "Transformers in remote sensing: A survey," 2022, *arXiv:2209.01206*.

[8] G. E. Hinton, S. Osindero, and Y.-W. Teh, "A fast learning algorithm for deep belief nets," *Neural Comput.*, vol. 18, no. 7, pp. 1527–1554, Jul. 2006.

[9] A. Pinkus, "Approximation theory of the MLP model in neural networks," *Acta Numerica*, vol. 8, pp. 143–195, Jan. 1999.

[10] Y. LeCun et al., "Backpropagation applied to handwritten zip code recognition," *Neural Comput.*, vol. 1, no. 4, pp. 541–551, Dec. 1989.

[11] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.

[12] E. B. Alexakis and C. Armenakis, "Evaluation of UNet and UNet++ architectures in high resolution image change detection applications," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 43, pp. 1507–1514, Aug. 2020.

[13] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[14] R. Jaturapitpornchai, M. Matsuoka, N. Kanemoto, S. Kuzuoka, R. Ito, and R. Nakamura, "Newly built construction detection in SAR images using deep learning," *Remote Sens.*, vol. 11, no. 12, p. 1444, Jun. 2019.

[15] Z. M. Hamdi, M. Brandmeier, and C. Straub, "Forest damage assessment using deep learning on high resolution remote sensing data," *Remote Sens.*, vol. 11, no. 17, p. 1976, Aug. 2019.

[16] D. Peng, Y. Zhang, and H. Guan, "End-to-end change detection for high resolution satellite images using improved UNet++," *Remote Sens.*, vol. 11, no. 11, p. 1382, Jun. 2019.

[17] H.-B. Bi, D. Lu, H.-H. Zhu, L.-N. Yang, and H.-P. Guan, "STA-Net: Spatial–temporal attention network for video salient object detection," *Int. J. Speech Technol.*, vol. 51, no. 6, pp. 3450–3459, Jun. 2021.

[18] T. Lei et al., "Difference enhancement and spatial–spectral nonlocal network for change detection in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4507013.

[19] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jan. 2022, Art. no. 5607514.

[20] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[21] C. Zhang, L. Wang, S. Cheng, and Y. Li, "SwinSUNet: Pure transformer network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5224713.

[22] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[23] H. Chen, C. Wu, B. Du, L. Zhang, and L. Wang, "Change detection in multisource VHR images via deep Siamese convolutional multiple-layers recurrent neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2848–2864, Apr. 2020.

[24] A. Singh, "Review article digital change detection techniques using remotely-sensed data," *Int. J. Remote Sens.*, vol. 10, no. 6, pp. 989–1003, Jun. 1989.

[25] W. Shi, M. Zhang, R. Zhang, S. Chen, and Z. Zhan, "Change detection based on artificial intelligence: State-of-the-art and challenges," *Remote Sens.*, vol. 12, no. 10, p. 1688, May 2020.

[26] S. Mishra, P. Shrivastava, and P. Dhurvey, "Change detection techniques in remote sensing: A review," *Int. J. Wireless Mobile Commun. Ind. Syst.*, vol. 4, no. 1, pp. 1–8, 2017.

[27] T. Bai et al., "Deep learning for change detection in remote sensing: A review," *Geo-Spatial Inf. Sci.*, pp. 1–27, Jun. 2022, doi: 10.1080/10095020.2022.2085633.

[28] K. Han et al., "A survey on visual transformer," 2020, *arXiv:2012.12556*.

[29] K. Subramanyam Kalyan, A. Rajasekharan, and S. Sangeetha, "AMMUS : A survey of transformer-based pretrained models in natural language processing," 2021, *arXiv:2108.05542*.

[30] S. Khan, M. Naseer, M. Hayat, S. W. Zamir, F. S. Khan, and M. Shah, "Transformers in vision: A survey," *ACM Comput. Surv.*, vol. 54, pp. 1–41, Jan. 2022.

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[31] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[32] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 213–229.

[33] R. Strudel, R. Garcia, I. Laptev, and C. Schmid, "Segmenter: Transformer for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7242–7252.

[34] Z. Niu, G. Zhong, and H. Yu, "A review on the attention mechanism of deep learning," *Neurocomputing*, vol. 452, pp. 48–62, Sep. 2021.

[35] M. H. Guo et al., "Attention mechanisms in computer vision: A survey," *Comput. Vis. Media*, vol. 8, no. 3, pp. 331–368, 2022.

[36] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[37] Z. Gao, J. Xie, Q. Wang, and P. Li, "Global second-order pooling convolutional networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3019–3028.

[38] Z. Yang, L. Zhu, Y. Wu, and Y. Yang, "Gated channel transformation for visual recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11791–11800.

[39] V. Mnih, N. Heess, A. Graves, and K. Kavukcuoglu, "Recurrent models of visual attention," in *Proc. Conf. Workshop Neural Inf. Process. Syst.*, 2014, pp. 1–9.

[40] X. Li, W. Wang, X. Hu, and J. Yang, "Selective kernel networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 510–519.

[41] J. Park, S. Woo, J.-Y. Lee, and I. So Kweon, "BAM: Bottleneck attention module," 2018, *arXiv:1807.06514*.

[42] S. Woo, J. Park, J. Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 3–19.

[43] F. Li et al., "Mask DINO: Towards a unified transformer-based framework for object detection and segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 3041–3050.

[44] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[45] Y. Ziang, Y. Zhao, and J. Dong, "Remote sensing image mining area change detection based on improved UNet Siamese network," *J. China Coal Soc.*, vol. 44, pp. 3773–3780, Oct. 2019.

[46] M. Zhang and W. Shi, "A feature difference convolutional neural network-based change detection method," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7232–7246, Oct. 2020.

[47] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.

[48] Z. Siyuan, S. Yonghong, Z. Yu, and Z. Yuanlin, "ECFNet: A Siamese network with fewer FPs and fewer FNs for change detection of remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[49] B. Hou, Q. Liu, H. Wang, and Y. Wang, "From W-Net to CDGAN: Bitemporal change detection via deep learning techniques," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1790–1802, Mar. 2020.



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